

ANX-PR/CL/001-01

LEARNING GUIDE

SUBJECT

103000393 - Challenges Of Accessible Computing For People With Functional Diversity

DEGREE PROGRAMME

10AK - Master Universitario En Software Y Sistemas

ACADEMIC YEAR & SEMESTER

2025/26 - Semester 1

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1. Description

1.1. Subject details

Name of the subject	103000393 - Challenges Of Accessible Computing For People With Functional Diversity
No of credits	4 ECTS
Type	Optional/elective
Academic year of the programme	First year
Semester of tuition	Semester 1
Tuition period	September-January
Tuition languages	English
Degree programme	10AK - Master Universitario en Software y Sistemas
Centre	10 - E.T.S. De Ingenieros Informáticos
Academic year	2025-26

2. Faculty

2.1. Faculty members with subject teaching role

Name and surname	Office/Room	Email	Tutoring hours *
Loic Antonio Martinez Normand	D2303	loic.mnormand@upm.es	Tu - 13:00 - 15:00 Th - 13:00 - 15:00 F - 13:00 - 15:00 Please confirm appointment via email. http://dlsiis.fi.upm.es/tutorias-2526

M. Carmen Suarez De Figueroa Baonza	D-2201	mdelcarmen.suarezdefigueroa@upm.es	M - 11:00 - 12:00 Tu - 12:00 - 14:00 Th - 14:00 - 17:00 Please confirm appointment via email.
Jose Luis Fuertes Castro (Subject coordinator)	D4307	joseluis.fuertes@upm.es	Tu - 17:00 - 20:00 W - 12:00 - 15:00 Please confirm appointment via email. http://dlsiis.fi.upm.es/tutorias-2526

* The tutoring schedule is indicative and subject to possible changes. Please check tutoring times with the faculty member in charge.

3. Skills and learning outcomes *

3.1. Skills to be learned

CEM1 - Identificar, a partir del estado de la cuestión, la presencia de problemas de investigación relacionados con la concepción, la construcción, el uso y la evaluación de sistemas sociotécnicos complejos que hagan un uso intensivo de software

CEM9 - Evaluar las tecnologías más innovadoras para la interacción persona-ordenador y juzgar de manera crítica las aportaciones a los problemas de investigación relacionados

CG12 - Comprensión amplia de las técnicas y métodos aplicables en una especialización concreta, así como de sus límites

CG13 - Apreciación de los límites del conocimiento actual y de la aplicación práctica de la tecnología más reciente.

CG2 - Que los estudiantes sean capaces de integrar conocimientos y enfrentarse a la complejidad de formular juicios a partir de una información que, siendo incompleta o limitada, incluya reflexiones sobre las responsabilidades sociales y éticas vinculadas a la aplicación de sus conocimientos y juicios.

CG4 - Que los estudiantes posean las habilidades de aprendizaje que les permitan continuar estudiando de un modo que habrá de ser en gran medida autodirigido o autónomo.

CGI23 - Capacidad de leer y comprender publicaciones dentro de su ámbito de estudio/investigación, así como su catalogación y valor científico

3.2. Learning outcomes

RA78 - Utilizar y definir métodos y herramientas para el diseño centrado en el usuario de productos TIC accesibles

RA76 - Conocer los principios de diseño para todos en relación con las TIC

RA75 - Conocer las ayudas técnicas y su interacción con TIC

RA77 - Utilizar y definir métodos y herramientas para la evaluación de la accesibilidad de productos TIC

RA79 - Utilizar normas técnicas de accesibilidad TIC y participar en su desarrollo

* The Learning Guides should reflect the Skills and Learning Outcomes in the same way as indicated in the Degree Verification Memory. For this reason, they have not been translated into English and appear in Spanish.

4. Brief description of the subject and syllabus

4.1. Brief description of the subject

This course provides a specialization about the accessibility of information and communication technologies (ICT) for persons with functional diversity (disability). It is mainly focused on current research issues in the field.

The course deals with an introduction to ICT accessibility concepts: functional diversity, design for all, user centred design, standards and the assessment of the accessibility degree of ICT products and services, cognitive accessibility and future trends in ICT accessibility.

After that, the students will work on current challenges in the field.

4.2. Syllabus

1. Functional diversity, accessibility and design for all
 - 1.1. Introduction
 - 1.2. Functional diversity
 - 1.3. Assistive products for ICT
 - 1.4. Principles of accessible design
 - 1.5. Introduction to Human-centred design
2. ICT accessibility standards
 - 2.1. Introduction to standards
 - 2.2. Relevant ICT accessibility standards
 - 2.3. Deeper study of one accessibility standard
 - 2.4. Conformity assessment
3. State of the art in ICT accessibility
 - 3.1. State of the art and future trends
4. Cognitive Accessibility
 - 4.1. Introduction to the Easy-to-Read Methodology

5. Schedule

5.1. Subject schedule*

Week	Type 1 activities	Type 2 activities	Distant / On-line	Assessment activities
1	Course introduction Duration: 00:20 Lecture Chapter 1: 1.1- Introduction Duration: 01:10 Lecture Chapter 1: 1.2- Functional diversity Duration: 00:30 Lecture			
2	Chapter 1: 1.2- Functional diversity Duration: 02:00 Lecture			Individual presentation of personas Individual presentation Progressive assessment Presential Duration: 00:20 Personas evaluation Other assessment Progressive assessment Presential Duration: 00:10
3	Chapter 1: 1.3- Assistive products Duration: 02:00 Lecture			
4	Chapter 1: 1.4- Principles of accessible design Duration: 01:00 Cooperative activities Chapter 1: 1.5- Introduction to human-centred design Duration: 00:30 Lecture Chapter 2: 2.1- Introduction to standards Duration: 00:30 Lecture			Individual presentation of principles of Design for All Individual presentation Progressive assessment Presential Duration: 00:30 Design for All evaluation Other assessment Progressive assessment Presential Duration: 00:10
5	Standards overview discussion Duration: 00:30 Cooperative activities Chapter 2: 2.2- Relevant ICT standards Duration: 00:45 Cooperative activities Chapter 2: 2.3- Deeper study of one accessibility standard Duration: 00:30 Lecture			Standard overview evaluation Other assessment Progressive assessment Presential Duration: 00:15 Test 1 Written test Progressive assessment Not Presential Duration: 00:30

	Explanation of exercise 1 Duration: 00:15 Lecture			
6	Chapter 2: 2.3- Deeper study of one accessibility standard Duration: 02:00 Cooperative activities			One accessibility standard evaluation Individual presentation Progressive assessment and Global Examination Presential Duration: 02:00
7	Chapter 2: 2.3- Deeper study of one accessibility standard Duration: 01:40 Cooperative activities Explanation of exercise 2 Duration: 00:20 Lecture			Delivery of exercise 1 Group work Progressive assessment Not Presential Duration: 00:00 One accessibility standard evaluation Individual presentation Progressive assessment and Global Examination Presential Duration: 02:00
8	Chapter 2: 2.4- Conformity assesment Duration: 01:45 Lecture Classroom tutoring. Exercise 2 Duration: 00:15 Additional activities			Conformity assesment evaluation Individual presentation Progressive assessment Presential Duration: 00:15
9	Chapter 3: 3.1- State of the art and future trends Duration: 01:45 Lecture Explanation of exercise 3 Duration: 00:15 Lecture			Delivery of exercise 2 Group work Progressive assessment Not Presential Duration: 00:00 State of the art in ICT accessibility evaluation Other assessment Progressive assessment Presential Duration: 00:10
10	Collective revision of exercise 2 Duration: 02:00 Cooperative activities			Participation in evaluation of exercise 2 Individual presentation Progressive assessment Presential Duration: 02:00
11	Chapter 4: Cognitive Accessibility Duration: 01:45 Lecture Explanation of exercise 4 Duration: 00:15 Lecture			Cognitive accessibility evaluation Other assessment Progressive assessment Presential Duration: 00:15
12	Chapter 4: Cognitive Accessibility Duration: 02:00 Lecture			Cognitive accessibility evaluation Other assessment Progressive assessment Presential Duration: 00:15

13	Group tutoring Duration: 02:00 Additional activities			
14	Classroom discussion about state of the art and future trends Duration: 02:00 Additional activities			Delivery of exercise 4 Group work Progressive assessment Not Presential Duration: 00:00 Presentation of exercise 3 Group presentation Progressive assessment and Global Examination Presential Duration: 02:00
15	Classroom discussion about state of the art and future trends Duration: 02:00 Additional activities			Presentation of exercise 3 Group presentation Progressive assessment and Global Examination Presential Duration: 02:00
16				Delivery of exercise 3 Group work Progressive assessment Not Presential Duration: 00:00
17				Delivery of exercise 1 Group work Global examination Not Presential Duration: 00:00 Delivery of exercise 2 Group work Global examination Not Presential Duration: 00:00 Delivery of exercise 3 Group work Global examination Presential Duration: 00:00 Delivery of exercise 4 Group work Global examination Not Presential Duration: 00:00 Test 1 Written test Global examination Not Presential Duration: 00:30 Test 2 Written test Progressive assessment and Global Examination Not Presential Duration: 00:30

Depending on the programme study plan, total values will be calculated according to the ECTS credit unit as 26/27 hours of student face-to-face contact and independent study time.

6. Activities and assessment criteria

6.1. Assessment activities

6.1.1. Assessment

Week	Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
2	Individual presentation of personas	Individual presentation	Face-to-face	00:20	1%	/ 10	CG2
2	Personas evaluation	Other assessment	Face-to-face	00:10	1%	/ 10	CG2
4	Individual presentation of principles of Design for All	Individual presentation	Face-to-face	00:30	2%	/ 10	CG12 CGI23 CEM9
4	Design for All evaluation	Other assessment	Face-to-face	00:10	1%	/ 10	CG4
5	Standard overview evaluation	Other assessment	Face-to-face	00:15	1%	/ 10	CG2 CGI23
5	Test 1	Written test	No Presential	00:30	10%	/ 10	CEM1 CG2 CG12 CEM9
6	One accessibility standard evaluation	Individual presentation	Face-to-face	02:00	5%	/ 10	CG4 CGI23
7	Delivery of exercise 1	Group work	No Presential	00:00	10%	/ 10	CEM1 CG2 CG4 CG12 CEM9
7	One accessibility standard evaluation	Individual presentation	Face-to-face	02:00	5%	/ 10	CG4 CGI23
8	Conformity assesment evaluation	Individual presentation	Face-to-face	00:15	1%	/ 10	CG4
9	Delivery of exercise 2	Group work	No Presential	00:00	15%	/ 10	CEM1 CG2 CG12 CEM9
9	State of the art in ICT accessibility evaluation	Other assessment	Face-to-face	00:10	1%	/ 10	CEM1 CG13 CEM9

10	Participation in evaluation of exercise 2	Individual presentation	Face-to-face	02:00	5%	/ 10	CG2 CG12 CEM9
11	Cognitive accessibility evaluation	Other assessment	Face-to-face	00:15	1%	/ 10	CEM9
12	Cognitive accessibility evaluation	Other assessment	Face-to-face	00:15	1%	/ 10	CEM9
14	Delivery of exercise 4	Group work	No Presential	00:00	10%	/ 10	CGI23 CEM9
14	Presentation of exercise 3	Group presentation	Face-to-face	02:00	5%	/ 10	CEM1 CG4 CG13 CGI23 CEM9
15	Presentation of exercise 3	Group presentation	Face-to-face	02:00	5%	/ 10	CEM1 CG4 CG13 CGI23 CEM9
16	Delivery of exercise 3	Group work	No Presential	00:00	10%	/ 10	CEM1 CG4 CG13 CGI23 CEM9
17	Test 2	Written test	No Presential	00:30	10%	/ 10	CEM1 CG2 CG4 CG12 CG13 CEM9

6.1.2. Global examination

Week	Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
6	One accessibility standard evaluation	Individual presentation	Face-to-face	02:00	5%	/ 10	CG4 CGI23
7	One accessibility standard evaluation	Individual presentation	Face-to-face	02:00	5%	/ 10	CG4 CGI23
14	Presentation of exercise 3	Group presentation	Face-to-face	02:00	5%	/ 10	CEM1 CG4 CG13 CGI23 CEM9
15	Presentation of exercise 3	Group presentation	Face-to-face	02:00	5%	/ 10	CEM1 CG4 CG13 CGI23 CEM9

17	Delivery of exercise 1	Group work	No Presential	00:00	10%	/ 10	CEM1 CG2 CG4 CG12 CEM9
17	Delivery of exercise 2	Group work	No Presential	00:00	20%	/ 10	CEM1 CG2 CG12 CEM9
17	Delivery of exercise 3	Group work	Face-to-face	00:00	15%	/ 10	CEM1 CG4 CG13 CG123 CEM9
17	Delivery of exercise 4	Group work	No Presential	00:00	15%	/ 10	CEM9
17	Test 1	Written test	No Presential	00:30	10%	/ 10	CEM1 CG2 CG12 CEM9
17	Test 2	Written test	No Presential	00:30	10%	/ 10	CEM1 CG2 CG4 CG12 CG13 CEM9

6.1.3. Referred (re-sit) examination

Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
Test 1	Written test	Face-to-face	00:30	10%	/ 10	CEM1 CG2 CG12 CEM9
Test 2	Written test	Face-to-face	00:30	10%	/ 10	CEM1 CG2 CG4 CG12 CG13 CEM9

Delivery of exercise 1	Individual work	Face-to-face	00:00	15%	/ 10	CEM1 CG2 CG4 CG12 CEM9
Delivery of exercise 2	Individual work	Face-to-face	00:00	20%	/ 10	CEM1 CG2 CG12 CEM9
Delivery of exercise 3	Individual work	Face-to-face	00:00	20%	/ 10	CEM1 CG4 CG13 CG123 CEM9
Presentation of exercise 3	Individual presentation	Face-to-face	02:00	10%	/ 10	CEM1 CG4 CG13 CG123 CEM9
Delivery of exercise 4	Individual work	Face-to-face	00:00	15%	/ 10	CEM9

6.2. Assessment criteria

The assessment of this module is divided into two parts: theory and practice. Both parts have to be passed in order to pass the module. The grades obtained in theory and practice are combined as described in the section on evaluation activities.

Theory

The theoretical part of the module contains different assessments: there will be two test-based assessments and there is going to be an assessment of the performance of the collaborative learning sessions that will be part of the study of accessibility standards ("One accessibility standard evaluation", that is not recoverable); there will be also short in-class evaluations during the semester (the grading shown here is approximate, as not all students will be able to participate in all progressive assessment activities).

Practical work

The practical work consists of 4 exercises:

- Exercise 1: a document containing change proposals for an accessibility standard.
- Exercise 2: an accessibility assessment of an ICT product, using the standard studied during collaborative learning.
- Exercise 3: state of the art on one topic related to ICT accessibility. Students will make a short presentation

(divided into two sessions) and deliver a report.

- Exercise 4: checking cognitive accessibility

Assessment procedure

The module will be assessed in a scale of 10 points, divided into theory and practical exercises. To pass the complete module it will be necessary to obtain a minimum of 3/10 point in theory and 3/10 points in the exercises.

a) Term evaluation

All the practical exercises are mandatory and will be graded according to the section on evaluation activities.

In addition, attendance, class participation and in-class activities will be graded for term evaluation.

b) Final evaluation

At the end of the term, there is the final evaluation. There will be two theory exams (the first one can be done by students having failed the first one). The four exercises have to be delivered in the same time period as the one defined for term evaluation, but they could be delivered again if not passed. The student will also have to attend the two collaborative sessions ("One accessibility standard evaluation") described that are not recoverable in the final evaluation. The exercise 3 presentations cannot be recoverables.

c) Extraordinary evaluation period

In the extraordinary evaluation period the theory tests not passed will be repeated and the pending exercises can be delivered again.

7. Teaching resources

7.1. Teaching resources for the subject

Name	Type	Notes
Don't make me think!: Revisited. A Common Sense Approach to Web Usability. 2014	Bibliography	Krug, S. New Riders, 3rd edition ISBN: 978-0321965516
The Principles of Universal Design. 1997	Bibliography	Connell, B.R.; Jones, M.; Mace, R.; Mueller, J.; Mullick, A.; Ostroff, E.; Sanford, J.; Steinfeld, E.; Story, M.; Vanderheiden, G. Version 2.0. North Carolina State University. https://design.ncsu.edu/research/center-for-universal-design/
Information technology -- User interface accessibility -- Part 1: User accessibility needs	Bibliography	International Organization for Standardization (ISO), International Electrotechnical Commission (IEC). ISO/IEC 29138-1:2018. (Technical report ISO/IEC TR 29138-1, 2009, can be accessed at http://jtc1access.org/TR29138.htm)
El modelo de la diversidad. La Bioética y los Derechos Humanos como herramientas para alcanzar la plena dignidad en la diversidad funcional. 2007	Bibliography	Palacios, A.; Romañach, J. Ediciones Diversitas, ISBN: 8496474402,
A Web for Everyone. Designing accessible user experiences. 2014	Bibliography	Horton, S.; Quesenbery, W. Rosenfeld
SIDAR	Web resource	Fundación Sidar - Acceso Universal: http://www.sidar.org , España
Accessibility Requirements for ICT products and services. V3.2.1. March 2021	Bibliography	EN 301 549. https://www.etsi.org/deliver/etsi_en/301500_301599/301549/03.02.01_60/en_301549v030201p.pdf

8. Other information

8.1. Other information about the subject

Exercises cannot be done just copying from other sources. Personal writing and analysis work by the student should be included (not third party or automatically generated reports are allowed). Failing to do this, implies plagiarism, which is not allowed at this University and will lead to not passing the exercise involved (grade will be 0).

Sustainable development goals (SDGs)

The goal of this course is to learn about enabling access of persons with disabilities to ICT, increasing their inclusion possibilities. Taking this into account, and considering the recommendations from the United Nations on the relationship between the SDGs and accessibility, this course is related to the following sustainable development goals:

- Goal 4 quality education - to ensure inclusive and equitable quality education and promote lifelong learning opportunities for all. In today's education, interactive learning systems are essential, and they need to be accessible and to be compatible with assistive products to enable the education of persons with disabilities.
- Goal 8 decent work and economy growth - to promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all. Today there are many job-related activities that rely on information and communication technology. This technology needs to be accessible and compatible with assistive products to enable inclusion in the workplace.
- Goal 10 reduced inequalities - to reduce inequality within and among countries. To increase inclusion of all persons in society, all interactive systems designed for citizen participation need to be accessible and be compatible with assistive products.